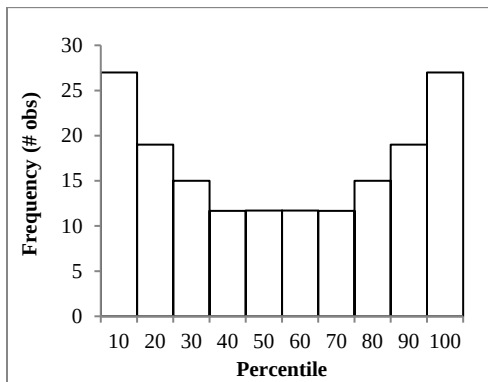


13. (3 points)

A stochastic loss reserving model estimates that ultimate losses have an expected value of \$70 million with a 90% confidence interval from \$60 million to \$84 million.

a. (1 point)

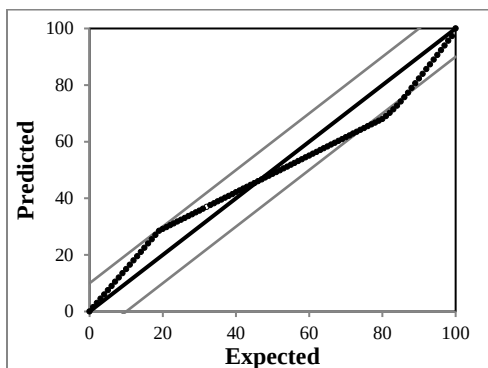
Assume the following histogram was produced during model validation:



Describe what is implied by the histogram about the accuracy of the expected value and the size of the confidence interval.

b. (1 point)

The following p-p plot was produced during model validation:



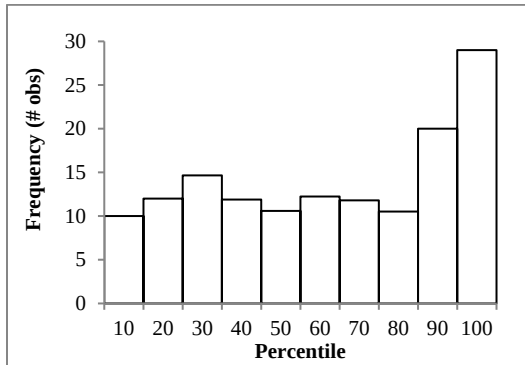
Describe what is implied by the plot about the accuracy of the expected value and the size of the confidence interval.

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(13 continued)

c. (1 point)

Assume the following histogram was produced during model validation:



Describe what is implied by the histogram about the accuracy of the expected value and the size of the confidence interval.

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EXAM 7 SPRING 2017 SAMPLE ANSWERS AND EXAMINER'S REPORT

QUESTION 13	
TOTAL POINT VALUE: 3	LEARNING OBJECTIVE(S): A9
SAMPLE ANSWERS	
Part a: 1 point	
<u>Sample 1</u> The histogram implies that our model is not heavy enough on the tails, since we are seeing far more than 10% of observed values on what should theoretically be the most extreme 10% (low and high) of our model. This would indicate that the model's confidence interval should be wider. However, the histogram is symmetrical, so at least our model appears to be unbiased relative to actual results – the mean (expected value) should be pretty accurate. <u>Sample 2</u> The bars for the highest and lowest percentile are taller than the rest. This indicates that the model is light tailed and is underestimating the size of the confidence interval. Since the histogram is symmetric, the model is equally light in both the left and right tails, so the expected value produced by the model is correct. <u>Sample 3</u> We want the histogram bars to be level, not so large at the extremes. The expected value is probably ok, but this is not a good model for this process. The model is too light in the tails. This means the model confidence interval is too small.	
Part b: 1 point	
<u>Sample 1</u> The backwards S shape indicates the model is heavy tailed and therefore the actual estimates are landing in the lowest + highest deciles less often than predicted. P-P plot appears symmetric indicating the expected value is okay but the variance is overstated. I would expect the min + max in the confidence interval to both more closer to the mean. <u>Sample 2</u> There are more observations at the middle percentiles than expected – suggests the model is heavy tailed. It does not appear asymmetrical or biased high or low so the expected value is reasonable. The confidence interval is too wide (large). <u>Sample 3</u> It shows that the model has heavy tail. Mean is reasonable = 70 mil. CI was overestimated. The actual results are not that extreme. CI should be (65, 75).	
Part c: 1 point	
<u>Sample 1</u> According to the histogram, there are many more observations in the 90th and 100th percentile buckets. Because of this the model is biased low. The model isn't predicting enough higher losses. Because of this, the expected value is too low. There is nothing to indicate that the model is heavy or light tailed, so the confidence interval (other than the expected value being off) seems ok.	

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Sample 2

Instead of the uniform Freq we expect, Actual values are higher in the right tail than expected, which means our model is biased low, and has a light right-tail. Expected value of model is too low, and upper confidence interval needs to be higher.

Sample 3

The histogram shows that predicted percentiles fall more in the higher percentile than in the middle and lower this indicates that model is biased low. This means that the expected value is not accurate, it should be higher. When expected value changes, confidence interval would change but the model does not indicate that the size of the interval is not accurate.

EXAMINER'S REPORT

Candidates were expected to understand what is implied about the accuracy of the expected value and the size of the confidence interval of a stochastic loss reserving model when the actual observations do not fit the model well.

Part a

Candidates were expected to discuss the points mentioned above when actual observations fall in the lowest and highest percentiles more often than the model predicts (model too light in the tails). The information provided to the candidates was a histogram showing the frequency of observations at each percentile.

Common errors include:

- Talking about predicted values instead of actual values when describing the histogram.
- Not being clear on whether the candidate was talking about the expected value/confidence interval of the predicted or actual distribution.
- Not recognizing the light tailed model situation, often confusing it with the heavy tailed model situation (both tails).

Part b

Candidates were expected to discuss the points mentioned above when actual observations fall in the middle percentiles more often than the model predicts (model too heavy in both tails). The information provided to the candidates was a p-p plot confronting expected and predicted percentiles.

Common errors include:

- Making comments about the model fit i.e. model is not a good fit because it is outside the K-S bands.
- Mixing up predicted and actual values when describing the graph.
- Not being clear on whether the candidate talked about the expected value/confidence interval of the predicted or actual distribution.
- Not recognizing the heavy tailed model situation, often confusing it with the light tailed model situation (both tails) or light left tail and heavy right tail situation.
- For the accuracy of the expected value, the candidates often justified the accuracy by the fact that 50th predicted percentile = 50th expected percentile, which is not

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a valid reason without stating the assumptions that both distributions follow a normal distribution (or any distribution where median = mean).

Part c

Candidates were expected to discuss the points mentioned above when actual observations fall in the highest percentiles more often than the model predicts (model too light in the right tail). The information provided to the candidates was a histogram showing the frequency of observations at each percentile.

Common errors include:

- The candidate often mixed up predicted and actual values when describing the graph.
- Not being clear on whether the candidate was talking about the expected value/confidence interval of the predicted or actual distribution.
- Not recognizing the biased low model situation, often confusing it with the biased high model situation (and reverse the expected value/confidence interval implications).
- Mentioning the confidence interval was accurate without justification. This answer was accepted when coupled with a valid justification (ex.: all percentiles except 90th and 100th are uniform).